

Ismail El Jamiy

PhD Student in Artificial Intelligence | Handwritten Text Recognition & Arabic NLP

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EDUCATION

University of Toulouse (IRIT, IMT, CLLE)	Toulouse – FRA
PhD in Artificial Intelligence	Nov. 2025 – Nov. 2028
Topic: Handwritten Text Recognition of medieval Arabic scientific manuscripts and development of dedicated LLMs.	
Supervisors: Farah Benamara, Guillaume Loizelet, Nabil Hathout Project: ATHAAR (ANR-22-EXES-0015)	
École Nationale Supérieure d'Arts et Métiers (ENSAM)	Casablanca – MAR
State Engineer's Degree, Artificial Intelligence and Computer Science Engineering	Sep. 2022 – Sep. 2025
Honors: Highest Distinction	
Integrated Preparatory Cycle, Mathematics, Physics and Engineering Sciences	Sep. 2020 – Jul. 2022

PUBLICATIONS

- [1] **I. El Jamiy**, G. Loizelet, F. Benamara, N. Hathout. *Reconnaissance automatique de manuscrits arabes scientifiques médiévaux : Création d'un jeu de données et évaluation de grands modèles de langues* [Automatic recognition of medieval Arabic scientific manuscripts: dataset creation and evaluation of large language models]. **CORIA-TALN 2026**, Nantes, France. [HAL: hal-05708366](#)
- [2] **I. El Jamiy**, A. Aissaoui, L. Chaari. *Few-Shot Medical Image Segmentation with Meta-Learning: An Empirical Study on Organ Segmentation*. **2025 Medical Technologies Congress (TIPTEKNO)**, Gazimağusa, Türkiye, 2025, pp. 1–4. DOI: [10.1109/TIPTEKNO68206.2025.11270127](#)

CONFERENCES, SCHOOLS & PRESENTATIONS

CORIA-TALN 2026 , Nantes, France: <i>oral presentation and poster</i>	Jun. – Jul. 2026
International Summer School The Paradigm Shift: From Rules to Models in NLP , Universidad de Alicante, Spain	Jun. 2026
EACL 2026 , 19th Conference of the European Chapter of the ACL, Rabat, Morocco	Mar. 2026
Second Arabic NLP School , co-located with EACL 2026, Rabat, Morocco	Mar. 2026

RESEARCH EXPERIENCE

Doctoral Researcher, ATHAAR project	Nov. 2025 – Present
Institut de Recherche en Informatique de Toulouse (IRIT)	Toulouse – FRA
Objective: Automatic recognition of al-Bīrūnī's <i>al-Qānūn al-Mas'ūdī</i> to support its critical edition.	
- Building an HTR ground-truth dataset from multiple manuscript witnesses held in international libraries (Calfa Vision, PAGE-XML). - Benchmarking vision-language models and LLMs in zero-shot and few-shot settings for medieval Arabic HTR. - Designing an error taxonomy for VLM-based Arabic HTR with an automatic rule-based error classifier.	
Research Intern in AI and Medical Imaging	Mar. 2025 – Sep. 2025
Institut de Recherche en Informatique de Toulouse (IRIT)	Toulouse – FRA
Objective: Medical image segmentation with limited data using meta-learning (MAML) and few-shot learning.	
- State-of-the-art review of segmentation under limited data and few-shot learning techniques. - Development of a U-Net-based segmentation model in a few-shot configuration on MRI and CT data.	

TEACHING & SUPERVISION

Teaching Assistant (32h, lab sessions)	2025 – 2026
Faculté Sciences et Ingénierie (FSI), University of Toulouse	Toulouse – FRA
Algorithms, Bachelor Year 1: 26h of lab sessions in Python (sorting algorithms, complexity analysis). Artificial Intelligence, Integrated preparatory program (CUPGE), Year 2: 6h of lab sessions.	
Supervision of an M2 research internship	Jul. 2026 – Sep. 2026
Institut de Mathématiques de Toulouse (IMT)	Toulouse – FRA
Defined the annotation strategy and wrote a detailed annotation guide for medieval Arabic scientific manuscripts.	

OTHER EXPERIENCE

Volunteer Computer Vision Developer

Aug. 2024 – Dec. 2024

Youth For Challenge Association

Casablanca – MAR

Objective: Intelligent platform facilitating communication for deaf and mute people through real-time translation of sign language into text.

- State-of-the-art review of hand detection and segmentation methods.
- Development of a CNN-based hand gesture recognition algorithm, deployed in a web interface with Flask and OpenCV.

Computer Vision Intern

Apr. 2024 – Jun. 2024

SYGMA.AI (InsurTech Company)

Casablanca – MAR

Objective: Segmentation of damaged vehicle areas after accidents to automatically assess insurance coverage.

- Collection and processing of car images (Python, Selenium, Scrapy).
- Implementation of a CNN for damaged vehicle classification with PyTorch.
- Development of a license plate verification algorithm combining CNNs and optical character recognition (OCR).

Academic Project: Marathon Photo Finder

Mar. 2024 – May 2024

 [Project Link](#)

ENSAM Casablanca

Objective: Help runners quickly find their race photos using image recognition.

- Implementation of a CNN-based person detection model.
- Development of an OCR system extracting bib numbers for runner re-identification.

AI and Image Processing Intern

Jul. 2023

Lesieur Cristal

Casablanca – MAR

Objective: Detection of personal protective equipment (PPE) in industrial videos to reduce operational risks and improve employee safety.

- Construction of a structured multi-class PPE dataset from diverse sources.
- Design of a CNN-based PPE detection and recognition algorithm.

SERVICE & OUTREACH

Google Developers Group On Campus, Cloud & DevOps Training Supervisor: 2 work-

Sep. 2024 – Mar. 2025

shops for 25 students and a conference on Infrastructure as Code.

MiaThon (national hackathon), Top 10 among 62 teams: travel-planning chatbot using

Jun. 2024

fine-tuned LLMs and RAG.

TECHNICAL SKILLS

- **Programming & Environment:** Python, LaTeX, Git, Linux, HPC (Jean Zay, SLURM).
- **Deep Learning:** PyTorch, HF Transformers, TensorFlow, Keras, Scikit-learn, OpenCV, Pillow.
- **LLMs & Agents:** LangChain, LangGraph, CrewAI.
- **HTR & Document Analysis:** Kraken, Calfa Vision, PAGE-XML, jiwer, vision-language models (Qwen2.5-VL, Gemini).
- **NLP:** spaCy, NLTK, pyarabic, CAMEL Tools.
- **Architectures & Methods:** CNN, U-Net, RNN, LSTM, Transformers, ViT, VLM, meta-learning (MAML), few-shot learning.

LANGUAGES

French: Bilingual (TCF C1, score 543) | **English:** Advanced | **Arabic:** Native language